Sentinel: The Checkout Trials

Gamified accessibility testing. A guide for judges.

Live app	https://sentinel-ten-ashy.vercel.app
Demo video	https://www.youtube.com/watch?v=aFqVHSjdlU8 (also in this zip)
Code (MIT)	https://github.com/FaatehHaneef/Sentinel
Sandbox test card	`4111 1111 1111 1111`, any future expiry date, any CVV

Try it in five minutes

1. Open the live app and press **Start your trial**. The shoppers walk into a line.
2. Grandma Iris is ticked. Tick **Tobin** too, and press **Start the trial**. Watch them walk the demo store. Each problem they meet is outlined on the page, and costs them coins and patience.
3. When they finish, press **See the full results**. You'll see who bought, the worst problems in plain words, and the money at risk each month.
4. Press **Try … fixes and run the trial again**. The suggested fixes are applied in your browser only, and the same shoppers walk again. Iris now gets through and pays in the Braintree sandbox.
5. Go back to the lobby and send **Cog** alone. Cog is an AI shopping agent: it can't buy until the store publishes machine-readable product data. Try its one fix, and Cog's agent decides to buy and places a sandbox order itself.
6. Open **The ledger** to see every problem across trials in an AG Grid table.

> Use **Pause** or **See as Iris** during a trial to stop or to see the page the way that shopper does. All seven shoppers can go at once with **Send everyone**, but that takes a few minutes.

Where PayPal and AI are used

- **Braintree sandbox (a PayPal service):** the demo store's real checkout. Line items with prices set on the server, protection against double payment, handled declines, and webhooks that are signature-checked.
- **Gemini:** writes each shopper's verdict from the measurements, suggests colours (used only if they pass the contrast maths), and runs Cog's buying agent.
- **Every number is measured by the browser** on the live page. The AI explains; it never makes up a measurement.

Things you might notice

- **"Wick's standard notes, written without AI."** Gemini runs on its free tier, which has a daily limit. When it's used up or Google is busy, Sentinel uses its own built-in notes and says so on screen, and Cog uses the ordinary checkout instead of its agent. Everything else works the same.
- **No PayPal button in the payment form.** PayPal accounts, including developer sandbox accounts, aren't available to developers in Pakistan, where Sentinel was built, so no PayPal account could be linked to the Braintree sandbox. Cards work. The PayPal button is wired in and turns on with one setting once an account is linked.
- **Records stay in your browser.** There are no accounts, so the ledger shows the trials run in your own browser.

Safety, enforced in code

- Trials run only on our demo store, or on sites whose owner added Sentinel's line of code with a signed key. On owners' sites the shoppers only look: they never type, pay or buy.
- Sandbox only: a live payments setting makes the app refuse to start.
- Only fixes from a fixed allow-list can ever be applied to a page.